Visual Question Answering with Joint CNN–Text Embeddings: An Empirical Study on the DAQUAR Dataset

Bassel Ali Mahmoud Ali

March 2021

***Abstract*—**Visual Question Answering (VQA) requires a system to jointly reason over an image and a natural-language question in order to produce an accurate natural-language answer. This paper presents a compact VQA baseline that combines bag-of-words text features with deep convolutional image features extracted from a pretrained CNN. The input question is represented as a one-hot vector, transformed into a dense representation through a learned embedding layer, and combined with the corresponding image feature vector; the joint representation is passed through a softmax classifier that predicts the answer as a multi-class label. The model is trained and evaluated on the DAtaset for QUestion Answering on Real-world images (DAQUAR), one of the earliest and smallest publicly available VQA benchmarks, consisting of 6,795 training and 5,673 testing question–answer pairs derived from the NYU-Depth V2 dataset. We systematically compare (i) bag-of-words versus recurrent (LSTM) question encoders, (ii) additive, multiplicative, and concatenative fusion strategies for combining visual and textual features, and (iii) the effect of dropout regularisation and bidirectional/stacked LSTM variants. The best configuration — bag-of-words question encoding, multiplicative fusion, and dropout regularisation — achieves 24.98% test accuracy, consistent with the expectation that simple architectures generalise better on small-scale datasets. We situate these results relative to more advanced VQA approaches, including attention-based and compositional models, and outline directions for future work on larger-scale datasets.

Keywords*—Visual Question Answering, Convolutional Neural Networks, Recurrent Neural Networks, Multimodal Fusion, DAQUAR, Deep Learning*

I. INTRODUCTION

A. Motivation

Visual Question Answering (VQA) is a task situated at the intersection of computer vision and natural language processing: given an image and a question about that image expressed in natural language, a system must produce an accurate natural-language answer. Computer vision addresses the problem of teaching machines to see — acquiring, processing, and interpreting visual information — while natural language processing (NLP) is concerned with enabling machines to read and generate human language. VQA requires both capabilities simultaneously and is therefore a more demanding test of multimodal reasoning than either field alone [1].

Unlike purely text-based question answering, image-based QA systems must ground their answers in visual content. Most contemporary image-QA models are neural [2]: a convolutional neural network (CNN) extracts a global image feature vector, a recurrent network such as an LSTM encodes the question, and the two representations are combined to infer an answer. This joint-embedding paradigm, closely related to image captioning [3], has become the standard baseline architecture against which more elaborate VQA methods are compared.

Relative to image captioning — where a system may select from many plausible descriptions — VQA is arguably harder, since the system must answer the specific question asked rather than generate any reasonable caption. Some questions are directly answerable from image content ("what is the colour of the boat?"), while others demand external or commonsense knowledge ("why is the baby crying?"). Building algorithms that approach human-level performance on this task is considered an important milestone on the road to more general machine intelligence.

B. Problem Statement

Formally, the VQA problem can be stated as follows: given an image and a natural-language question about that image, construct a system that produces a natural-language answer grounded in the visual content of the image. Figure 1 illustrates the task: given a photograph and the question "What is the mustache made of?", the system must correctly output "bananas."

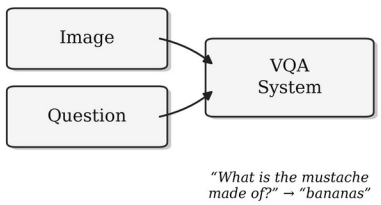


Figure 1. Schematic view of the VQA task: an image and a natural-language question are jointly mapped to a natural-language answer.

More generally, an ideal VQA system should be able to answer arbitrary questions about an arbitrary image

with reference to that image's content — for example, counting objects ("how many books are on the shelf?"), identifying interactions ("what is the cat playing with?"), or describing attributes ("what colour are the books?").

C. Objectives and Applications

VQA has a number of practical applications that motivate continued research:

1. Accessibility. A VQA system can describe the content of an image on the web or on social media to blind and visually impaired users [4].

2. Image retrieval. VQA can be integrated into retrieval systems, allowing databases of images to be filtered or queried using natural-language questions rather than fixed tags — with direct applications in social media and e-commerce.

D. Pipeline Overview

The architecture explored in this work follows the standard joint-embedding formulation and consists of three stages, illustrated in Figure 2 for the example question "are these people family?": (i) visual features are extracted from the input image using a pretrained CNN; (ii) the question is converted into a one-hot vector and mapped to a word/question feature vector; and (iii) the two feature vectors are fused and passed through a softmax classifier over the answer vocabulary.

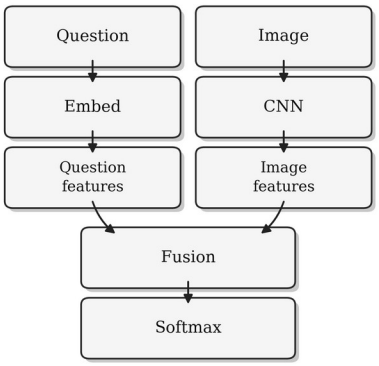


Figure 2. Overview of the joint-embedding VQA pipeline: CNN-derived image features and word features are fused and classified via softmax.

Three design choices govern this pipeline and form the basis of the experiments in Section IV: (a) the method used to combine the image and question feature vectors (addition, multiplication, or concatenation); (b) the method used to encode the question (bag-of-words versus a recurrent encoder such as an LSTM); and (c) the convolutional architecture used to extract image features (e.g. VGGNet or GoogLeNet).

II. THEORETICAL BACKGROUND

A. Convolutional Neural Networks

Convolutional Neural Networks (CNNs) [5] are feedforward networks specialised for grid-structured data such as images. Unlike a fully-connected network — in which the number of parameters scales with the product of the input dimensions and quickly becomes intractable for realistic image sizes — a CNN constrains its architecture by arranging neurons in three dimensions (width, height, depth) and enforcing local, weight-shared connectivity. A typical CNN is built from three layer types: convolutional layers, which compute local dot products between learned filters and small input regions; pooling layers, which perform spatial downsampling; and fully-connected layers, which compute the final class scores. Layers with parameters (CONV, FC) are trained by gradient descent, while non-parametric layers (ReLU, POOL) implement fixed functions. CNNs are used throughout this work as fixed feature extractors for the visual branch of the VQA pipeline (Section III-C1).

B. Recurrent Neural Networks and LSTMs

Recurrent Neural Networks (RNNs) [6] are designed for sequential data: unlike feedforward networks, they maintain a hidden state that is updated at each time step, allowing previous outputs to influence subsequent computation. This makes RNNs a natural choice for encoding variable-length natural-language questions. However, plain RNNs struggle with long-range dependencies, since the influence of an early input on a later prediction attenuates over the sequence.

Long Short-Term Memory (LSTM) networks were introduced by Hochreiter and Schmidhuber to address this long-dependency problem by augmenting the hidden state with a memory cell governed by input, output, and forget gates that regulate the flow of information [25]. Given input x at time t and previous hidden state h, the forward pass of an LSTM cell with a forget gate is:

$$f_t = \sigma_g(W_f x_t + U_f h_{t-1} + b_f)$$

$$i_t = \sigma_g(W_i x_t + U_i h_{t-1} + b_i)$$

$$o_t = \sigma_g(W_o x_t + U_o h_{t-1} + b_o)$$

$$\tilde{c}_t = \sigma_c(W_c x_t + U_c h_{t-1} + b_c)$$

$$c_t = f_t \circ c_{t-1} + i_t \circ \tilde{c}_t$$

$$h_t = o_t \circ \sigma_h(c_t)$$

where f_t, i_t and o_t denote the forget, input, and output gates respectively, c_t is the cell state, h_t is the hidden state, σ denotes the sigmoid (gate) or tanh (candidate/output) nonlinearity, and $\circ$ denotes the Hadamard (element-wise) product.

Bidirectional RNNs (BRNNs) [7] extend this formulation by processing the input sequence in both temporal directions with two independent hidden-state paths, allowing predictions to draw on both past and future context — useful whenever the entire input sequence is available ahead of time, as is the case for a question in VQA.

Gated Recurrent Units (GRUs) [26] are a simplified alternative to the LSTM that merge the cell and hidden states and reduce the gating mechanism to two gates — a reset gate r_t and an update gate z_t:

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$$

$$h'_t = \tanh(W \cdot [r_t * h_{t-1}, x_t])$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * h'_t$$

Here h'_t denotes the candidate hidden state (equivalently written $\tilde{h}_t$ in some notations). GRUs are computationally cheaper than LSTMs while offering comparable performance on many sequence tasks, which motivates their inclusion as a candidate question-encoder architecture in this study.

C. Dropout Regularisation

Dropout [8] is a regularisation technique that approximates training a large ensemble of differently-structured networks in parallel. At each training step, a randomly-selected fraction p of a layer's units are temporarily set to zero, forcing the remaining units to be more independently useful and reducing co-adaptation, whereby multiple neurons converge on redundant representations. Dropout is applied only to hidden (and optionally input) layers, never to the output layer, and is disabled at test time. In this study, dropout is applied to the fused image–question representation prior to the final softmax classifier.

D. Batch Normalisation

Training deep networks is complicated by internal covariate shift — the tendency of the distribution of layer inputs to drift as the parameters of preceding layers are updated during training. Batch normalisation [9] addresses this by standardising the activations of a layer over each mini-batch. Given a mini-batch $B = \{x_1 \ldots x_m\}$, the per-dimension mean and variance are computed as:

$$\mu_B = (1/m) \sum_{i=1}^{m} x_i$$

$$\sigma_B^2 = (1/m) \sum_{i=1}^{m} (x_i - \mu_B)^2$$

Each activation is then standardised and re-scaled using two learned parameters, γ (scale) and β (shift), which allow the network to recover the original representation if that is optimal:

$$\hat{x}_i^{(k)} = (x_i^{(k)} - \mu_B^{(k)}) / \sqrt{(\sigma_B^{(k)2} + \varepsilon)}$$

$$y_i^{(k)} = \gamma^{(k)} \hat{x}_i^{(k)} + \beta^{(k)}$$

Batch normalisation stabilises and accelerates training, allows for larger learning rates, and provides a mild regularising effect. While batch normalisation is standard practice for deeper CNN backbones, this study relies primarily on dropout for regularisation, since the classifier network trained on top of the fixed CNN features is comparatively shallow.

E. Transfer Learning

Transfer learning [10] refers to the reuse of knowledge acquired while solving one problem to accelerate or improve performance on a different but related problem. In computer vision, it is standard practice to pretrain a CNN on a large-scale dataset such as ImageNet (1.2 million images, 1,000 categories) [19] and then reuse it for a target task, either (a) as a fixed feature extractor, where the final classification layer is discarded and the remaining activations are used as generic image descriptors, or (b) via fine-tuning, where the pretrained weights are used as an initialisation and updated further by continued back-propagation on the target dataset. Because early CNN layers learn generic features (edges, colour blobs) while later layers become increasingly dataset-specific, the appropriate transfer-learning strategy depends on the size of the target dataset and its similarity to the source domain. Given the small size of DAQUAR, this study uses pretrained CNNs strictly as fixed feature extractors (Section III-C1), consistent with standard practice for small, dissimilar target datasets.

F. Deep Learning, Computer Vision, and NLP in Context

Deep learning [11] refers to neural network models with many stacked layers that learn hierarchical representations directly from raw data, without hand-engineered features. It underlies state-of-the-art systems in image recognition, captioning, question answering, and speech and language processing, and is the common substrate for both the vision and language components of the VQA pipeline studied here. In computer vision [15], CNN-based models now exceed human accuracy on many object-recognition benchmarks; in NLP [14], word-embedding methods and recurrent or sequence-to-sequence architectures allow models to capture both syntactic structure and semantic relationships between words — for example, embedding words such that the vector offset between "king" and "queen" approximates that between "man" and "woman." Both branches converge directly in VQA, which requires a single system to perform image understanding, language understanding, and cross-modal reasoning simultaneously.

III. METHODOLOGY

A. Dataset Landscape

Since 2014, several major datasets for VQA have been publicly released, including DAQUAR, COCO-QA, the VQA Dataset, FM-IQA, Visual7W, and Visual Genome [16]. With the exception of DAQUAR, all of these use images from the Microsoft COCO dataset (328,000 images, 91 object categories, over 2 million labelled instances). An ideal VQA dataset must be large enough to capture the variability of real-world questions, images, and concepts, and must resist exploitable biases that would allow a model to perform well without genuinely solving the underlying task.

B. The DAQUAR Dataset

This study uses the DAtaset for QUestion Answering on Real-world images (DAQUAR) [17], the first major VQA dataset released and one of the smallest currently available. DAQUAR consists of 6,795 training and 5,673 testing question–answer pairs derived from images in the NYU-Depth V2 indoor-scene dataset. A reduced configuration, DAQUAR-37, restricts the object vocabulary to 37 categories (3,825 training / 297 testing pairs); an alternative evaluation protocol, DAQUAR-consensus, incorporates additional human-collected ground-truth answers. DAQUAR contains exclusively indoor scenes with significant clutter and, in places, extreme lighting, which constrains question variety and makes the dataset genuinely difficult — even human annotators achieve only 50.2% accuracy on the full dataset. Its small size makes DAQUAR unsuitable for training very large models, but well suited to the systematic, compute-constrained baseline comparison undertaken in this study.

C. Joint Embedding Architecture

Visual Question Answering is a multimodal learning problem: given an image and a natural-language question about that image, the system must produce an answer. VQA models are often complex and prone to poor generalisation on previously unseen questions, and data collection is comparatively expensive, which motivates baseline architectures that are simple enough to train reliably on small datasets while still capturing the core cross-modal reasoning challenge.

The most common approach to VQA transfers both the image and the question into a shared embedding space — the joint embedding approach, first explored in the context of image captioning [18]. In its most basic form, this proceeds in three stages: (i) extract a useful representation of the image and of the question; (ii) combine the two representations while retaining information from both modalities; and (iii) use the combined representation to determine an output answer.

1) Image and Question Feature Extraction:

Image features are extracted using a CNN pretrained for object recognition on ImageNet [19], with the final classification layer removed so that the network acts as a fixed feature extractor, yielding a flattened 1×N vector representation of the image. Commonly used backbones include VGGNet [20], GoogLeNet [21], and ResNet [22]; this study uses VGGNet and GoogLeNet variants, whose architecture is illustrated in Figure 3.

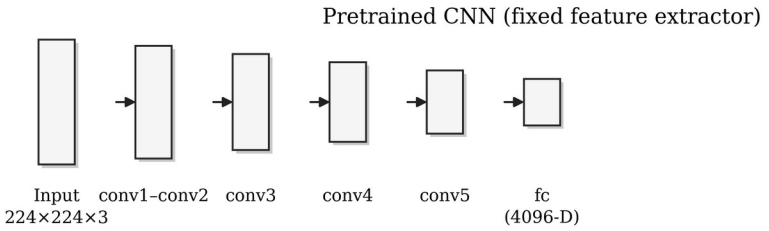


Figure 3. Schematic of the CNN feature-extraction pipeline: successive convolutional blocks reduce spatial resolution while increasing feature depth, yielding a 4096-D fully-connected (fc) activation vector used as the fixed visual feature representation.

Question features are obtained either with a bag-of-words representation or with a recurrent encoder. Many joint-embedding models pair a pretrained word embedding (e.g. Word2Vec [23] or GLoVe [24]) with an RNN — typically an LSTM [25] or GRU [26] — that consumes the embedded question token by token; the RNN's final internal state then serves as a featurisation of the entire question. RNN-based encoders have been criticised for being slow to train and difficult to tune, and some studies find convolutional architectures competitive with or superior to RNNs for sequence modelling [27], [28]. In this study, we let the network learn the word embedding from scratch, and compare two encoders: (a) a simple bag-of-words sum over learned word embeddings, and (b) an LSTM encoder. Consistent with the small size of DAQUAR, bag-of-words gives better results because of its relative simplicity (see Section IV).

2) Multimodal Feature Fusion:

Given a visual feature vector and a question feature vector, the two must be combined into a single joint representation. Common fusion operators include:

- Element-wise addition of the two vectors.
- Element-wise multiplication (Hadamard product) of the two vectors.
- Concatenation (simple adjoining) of the two vectors.
- Bilinear pooling: the outer product of the two vectors, capturing full second-order interactions between all elements.

Bilinear pooling models rich cross-modal interactions but is computationally expensive, since the outer product scales with the product of the two feature dimensionalities; practical systems therefore typically rely on compact approximations, such as Multimodal Compact Bilinear pooling (MCB) [29], which projects both representations into a lower-dimensional space and combines them via an element-wise product in the Fourier domain. Other approaches, such as Multimodal Residual Learning [30], use residual blocks to learn an optimal joint representation directly. For computational tractability on DAQUAR, this study restricts its comparison to the three basic linear fusion operators — addition, multiplication, and concatenation — detailed in Section IV.

Given the fused representation, the most common way to produce an answer is to treat VQA as a classification problem: the top-n answers observed in the training data define a fixed set of classes, and a multilayer perceptron (typically ending in a softmax layer of size n) predicts among them. This is computationally inexpensive and works well for datasets such as DAQUAR, whose answers are predominantly short (often single words), but it cannot represent free-form or previously unseen answers and cannot compose novel combinations of known concepts. Treating VQA instead as a sequence-generation problem, with a decoder RNN producing the answer token by token from a shared "thought vector," addresses this limitation at the cost of additional model and training complexity; most DAQUAR-scale studies, including this one, adopt the classification formulation.

D. Model Architecture Used in This Study

The final architecture used for our experiments is shown in Figure 4. The question (padded/truncated to 30 tokens) is passed through an embedding layer, summed into a single vector (bag-of-words), and projected through two dense layers to a 100-dimensional question representation with batch normalisation. In parallel, the 1000-dimensional CNN image feature vector is projected to the same 100-dimensional space. The two 100-D vectors are then fused via element-wise multiplication, passed through a dense layer, regularised with dropout, and finally mapped by a softmax layer over 503 candidate answer classes.

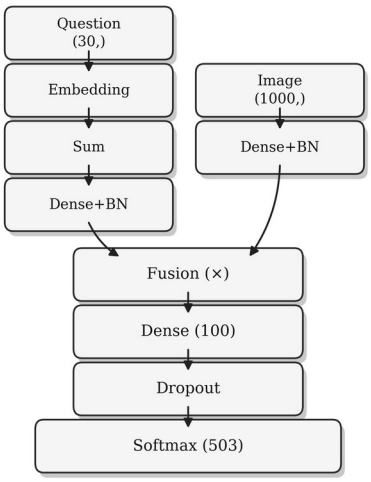


Figure 4. Computation graph of the trained model: question embedding and image feature branches are each projected to a 100-D space, fused by element-wise multiplication, regularised with dropout, and classified over 503 answer classes.

IV. EXPERIMENTS AND RESULTS

All experiments are trained on the DAQUAR training split and evaluated on the held-out DAQUAR test split, using top-1 accuracy on the first answer word as the evaluation metric. We evaluate the effect of four design choices, holding the remainder of the architecture fixed at the configuration described in Section III-D.

A. Question Encoding: Bag-of-Words vs. LSTM

Table I compares a simple bag-of-words question encoder against a single-layer LSTM encoder, with all other settings fixed.

TABLE I
QUESTION ENCODING: BAG-OF-WORDS VS. LSTM

Question Encoder	Test Accuracy
Bag-of-Words	**0.2138**
LSTM	**0.1847**

Bag-of-words outperforms the LSTM encoder. This is consistent with the small size of DAQUAR: an LSTM introduces substantially more trainable parameters and is more prone to overfitting a dataset of this scale, whereas the simpler bag-of-words representation generalises more reliably.

B. Multimodal Fusion Strategy

Table II compares concatenation, addition, and multiplication as the fusion operator between the image and question feature vectors.

TABLE II
MULTIMODAL FUSION STRATEGY COMPARISON

Fusion Method	Test Accuracy
Concatenation	**0.2101**
Addition	**0.2138**
Multiplication	**0.2329**

Multiplicative (Hadamard) fusion performs best, suggesting that an element-wise gating interaction between visual and textual features — where each modality can amplify or suppress the other — is more informative for this task than either simple summation or the higher-dimensional but less-structured concatenated representation.

C. Effect of Dropout

Table III shows the effect of adding a dropout layer after the fused representation, using the best-performing configuration from Sections IV-A–IV-B (bag-of-words, multiplicative fusion).

TABLE III
EFFECT OF DROPOUT REGULARISATION

Configuration	Test Accuracy
With Dropout	**0.2498**
Without Dropout	**0.2138**

Dropout yields a substantial improvement (+3.6 points), confirming its value as a regulariser on a dataset as small as DAQUAR, where overfitting is the primary obstacle to generalisation.

D. Recurrent Architecture Variants

For completeness, Table IV compares two more elaborate recurrent question encoders — a bidirectional LSTM and a two-layer stacked LSTM — against one another.

TABLE IV
RECURRENT ARCHITECTURE VARIANTS

Recurrent Variant	Test Accuracy
Bidirectional LSTM	**0.2260**
2-Layer LSTM	**0.2202**

The bidirectional and stacked LSTM variants perform similarly to one another, and both improve on the single-layer LSTM baseline in Table I, indicating that additional context (from bidirectionality) or depth (from stacking) partially offsets the LSTM's tendency

to overfit — though neither variant surpasses the simple bag-of-words encoder combined with dropout.

E. Best Configuration and Discussion

The best overall configuration — bag-of-words question encoding, multiplicative image–question fusion, and dropout regularisation — achieves 24.98% test accuracy on DAQUAR. This is a decent result for such a small and challenging dataset, given that even human annotators achieve only 50.2% accuracy on DAQUAR's full evaluation protocol. The consistent pattern across all four experiments — simpler components and explicit regularisation outperforming more expressive but higher-capacity alternatives — underscores that dataset scale, not architectural sophistication, is the binding constraint in this setting.

V. DISCUSSION: RELATED AND EXTENDED APPROACHES

VQA is a central research problem in the intersection of computer vision and NLP, requiring more of a system than single-purpose tasks such as object recognition or detection: an algorithm able to answer arbitrary questions about arbitrary images would represent a significant milestone toward general visual intelligence, and VQA is often proposed as a necessary component of any visual Turing test. The baseline explored in this paper is deliberately simple; this section discusses more advanced techniques that address its principal limitations.

A. Attention Mechanisms

The joint-embedding baseline in Section III uses the entire image representation regardless of the question asked, which is somewhat counter to how a human would approach the task: cognitive-psychology findings suggest that human vision does not process a scene as a uniformly detailed, picture-like representation [31], but instead attends selectively to regions relevant to the task at hand. Attention mechanisms bring this idea to VQA models [32] by computing a weighted distribution over spatial regions of the image, conditioned on the question, and combining only the attended (image-weighted) features with the question representation before classification (Figure 5).

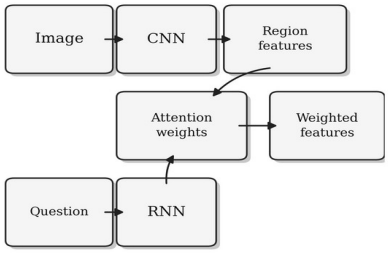


Figure 5. Attention-based VQA: region-specific CNN features are weighted by a question-conditioned attention distribution before being passed to the classifier or decoder.

Beyond improving accuracy, attention offers a degree of explainability, since the attention weights can be visualised as a heat map over the input image, showing which regions the network relied upon when producing its answer. A key design question for attention-based VQA is how to represent local image features — common choices include a uniform m×n grid of CNN features taken before the final pooling/flattening operation, non-uniform proposed regions encoded individually, or bounding boxes; datasets such as VQA-HAT [38], in which human annotators mark the image regions they used to answer each question, provide a way to train or benchmark attention mechanisms directly against human gaze/attention behaviour.

Stacked Attention Networks (SANs) [33] extend this idea by querying the image multiple times: an initial attention layer identifies broadly relevant regions, and subsequent layers progressively narrow attention to the most relevant region(s) needed to answer the question, refining the answer through successive reasoning steps.

B. Compositional Models

A limitation shared by CNN/RNN-based joint-embedding methods, including the baseline studied here, is that they are architecturally monolithic: the same computation is applied regardless of a question's complexity, even though "is this a car?" and "why is the man to the right of the third chair holding a hammer in his left hand?" clearly demand very different reasoning. Compositional models address this by assembling question-specific computation graphs from smaller, reusable modules. Neural Module Networks (NMN) [34] define a library of modules, each specialised for one sub-task (e.g. attending to a region, classifying based on an attended region), and use a parser to convert the question into a symbolic tree whose nodes are mapped onto specific modules — so the network's architecture is itself a

function of the question. Dynamic Memory Networks (DMN) [35], later adapted to VQA [36], instead maintain an explicit memory of "facts" extracted from both the question and the image (via bidirectional GRUs) and use an attention-based episodic memory module to iterate over these facts multiple times before producing an answer.

C. External Knowledge-Base Approaches

Some questions require knowledge that is not present in the image itself — for instance, recognising the Eiffel Tower and answering "which city is pictured here?" requires knowing that the structure is located in Paris. Because standard joint-embedding methods can only draw on information present in the training set, several approaches augment VQA models with external knowledge bases such as DBpedia [37], for example by extracting candidate image attributes with a CNN, querying an external database with those attributes, and combining the retrieved information with a CNN-derived caption representation before decoding an answer with an LSTM.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

The recent rise of deep learning methods — particularly convolutional and recurrent neural networks — has substantially advanced a wide range of computer vision and NLP tasks, in several cases matching or exceeding human performance. Tasks such as object and scene classification demonstrate reliable holistic and regional image recognition, while tasks such as caption generation and cross-modal retrieval demonstrate that deep learning can successfully bridge the gap between images and language. Visual Question Answering extends this boundary further still, requiring a system to understand a natural-language question, locate or classify the entities and relations that question refers to within an image, and generate an appropriate natural-language answer — potentially drawing on all of the capabilities above simultaneously.

In this project, we built a visual question answering system on the DAQUAR dataset, implementing and evaluating a simple joint-embedding baseline, and comparing it against more advanced published approaches, including compositional models and stacked attention networks. Our experiments show that, on a dataset as small as DAQUAR, simplicity and explicit regularisation — bag-of-words question encoding, multiplicative fusion, and dropout — outperform more expressive components such as LSTM question encoders, achieving a best test accuracy of 24.98%.

B. Future Work

This study used a single family of methods — joint embedding models — evaluated on a small dataset, and found that model capacity is constrained more by data scale than by architectural choice. Natural extensions of this work include: (i) training and evaluating the same family of models on larger-scale VQA datasets, such as the VQA Dataset or Visual Genome, where higher-capacity encoders (LSTM, bidirectional/stacked LSTM) would be expected to close or reverse the gap observed here; (ii) incorporating attention mechanisms, which have been shown to improve accuracy substantially on larger datasets and would let the model condition on question-relevant image regions rather than a single global feature vector; and (iii) exploring compositional architectures such as Neural Module Networks, which may generalise better to the wide range of question complexities present in larger, more realistic VQA benchmarks.

REFERENCES

[1] Sharma, Yashvardhan, and Sahil Gupta. "Deep learning approaches for question answering system." Procedia Computer Science 132 (2018): 785–794.

[2] Zhou, Bolei, et al. "Simple baseline for visual question answering." arXiv preprint arXiv:1512.02167 (2015).

[3] Vinyals, Oriol, et al. "Show and tell: Lessons learned from the 2015 MSCOCO image captioning challenge." IEEE Transactions on Pattern Analysis and Machine Intelligence 39.4 (2016): 652–663.

[4] Gurari, Danna, et al. "VizWiz grand challenge: Answering visual questions from blind people." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2018.

[5] Albawi, Saad, Tareq Abed Mohammed, and Saad Al-Zawi. "Understanding of a convolutional neural network." 2017 International Conference on Engineering and Technology (ICET). IEEE, 2017.

[6] Medsker, Larry R., and L. C. Jain. "Recurrent neural networks." Design and Applications 5 (2001).

[7] Schuster, Mike, and Kuldip K. Paliwal. "Bidirectional recurrent neural networks." IEEE Transactions on Signal Processing 45.11 (1997): 2673–2681.

[8] Srivastava, Nitish, et al. "Dropout: a simple way to prevent neural networks from overfitting." Journal of Machine Learning Research 15.1 (2014): 1929–1958.

[9] Bjorck, J., Gomes, C., Selman, B., & Weinberger, K. Q. “Understanding batch normalization.” arXiv preprint arXiv:1806.02375 (2018).

[10] Torrey, Lisa, and Jude Shavlik. “Transfer learning.” Handbook of Research on Machine Learning Applications and Trends: Algorithms, Methods, and Techniques. IGI Global, 2010. 242–264.

[11] LeCun, Yann, Yoshua Bengio, and Geoffrey Hinton. “Deep learning.” Nature 521.7553 (2015): 436–444.

[12] Sternberg, S. R., Dargel, W., Lennington, J. W., & Voiles, T. U.S. Patent No. 6,763,148. Washington, DC: U.S. Patent and Trademark Office, 2004.

[13] Zhou, Li, and Kevin Small. “Multi-domain dialogue state tracking as dynamic knowledge graph enhanced question answering.” arXiv preprint arXiv:1911.06192 (2019).

[14] Sarikaya, Ruhi, Geoffrey E. Hinton, and Anoop Deoras. “Application of deep belief networks for natural language understanding.” IEEE/ACM Transactions on Audio, Speech, and Language Processing 22.4 (2014): 778–784.

[15] Szeliski, Richard. Computer Vision: Algorithms and Applications. Springer Science & Business Media, 2010.

[16] Wu, Qi, et al. “Visual question answering: A survey of methods and datasets.” Computer Vision and Image Understanding 163 (2017): 21–40.

[17] Malinowski, Mateusz, and Mario Fritz. “Towards a visual Turing challenge.” arXiv preprint arXiv:1410.8027 (2014).

[18] Vinyals, O., Toshev, A., Bengio, S., & Erhan, D. “Show and tell: A neural image caption generator.” CoRR abs/1411.4555 (2014).

[19] Russakovsky, O., Deng, J., Su, H., et al. “ImageNet large scale visual recognition challenge.” International Journal of Computer Vision 115.3 (2015): 211–252.

[20] Simonyan, K., and Zisserman, A. “Very deep convolutional networks for large-scale image recognition.” arXiv:1409.1556 (2014).

[21] Szegedy, C., Liu, W., Jia, Y., et al. “Going deeper with convolutions.” CoRR abs/1409.4842 (2014).

[22] He, K., Zhang, X., Ren, S., & Sun, J. “Deep residual learning for image recognition.” CoRR abs/1512.03385 (2015).

[23] Mikolov, T., Corrado, G., Chen, K., & Dean, J. “Efficient estimation of word representations in vector space.” 2013, pp. 1–12.

[24] Pennington, J., Socher, R., & Manning, C. “GloVe: Global vectors for word representation.” Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP). Doha, Qatar: ACL, 2014, pp. 1532–1543.

[25] Hochreiter, S., and Schmidhuber, J. “Long short-term memory.” Neural Computation 9 (1997): 1735–1780.

[26] Cho, K., van Merriënboer, B., Gulcehre, C., et al. “Learning phrase representations using RNN encoder-decoder for statistical machine translation.” CoRR abs/1406.1078 (2014).

[27] Ma, L., Lu, Z., & Li, H. “Learning to answer questions from image using convolutional neural network.” 2015.

[28] Wang, Z., and Ji, S. “Learning convolutional text representations for visual question answering.” 2017.

[29] Fukui, A., Park, D. H., Yang, D., et al. “Multimodal compact bilinear pooling for visual question answering and visual grounding.” 2016.

[30] Kim, J., Lee, S., Kwak, D., et al. “Multimodal residual learning for visual QA.” CoRR abs/1606.01455 (2016).

[31] Rensink, R. “The dynamic representation of scenes.” Visual Cognition 7 (2000): 17–42.

[32] Kafle, K., and Kanan, C. “Visual question answering: Datasets, algorithms, and future challenges.” Computer Vision and Image Understanding, 2016.

[33] Yang, Z., He, X., Gao, J., Deng, L., & Smola, A. J. “Stacked attention networks for image question answering.” CoRR abs/1511.02274 (2015).

[34] Andreas, J., Rohrbach, M., Darrell, T., & Klein, D. “Deep compositional question answering with neural module networks.” CoRR abs/1511.02799 (2015).

[35] Kumar, A., Irsoy, O., Su, J., et al. “Ask me anything: Dynamic memory networks for natural language processing.” CoRR abs/1506.07285 (2015).

[36] Xiong, C., Merity, S., & Socher, R. “Dynamic memory networks for visual and textual question answering.” CoRR abs/1603.01417 (2016).

[37] Wu, Q., Wang, P., Shen, C., Dick, A., & van den Hengel, A. “Ask me anything: Free-form visual question answering based on knowledge from external sources.” 2015.

[38] Das, A., Agrawal, H., Zitnick, L., Parikh, D., & Batra, D. “Human attention in visual question answering: Do humans and deep networks look at the same regions?” Computer Vision and Image Understanding 163 (2017): 90–100.